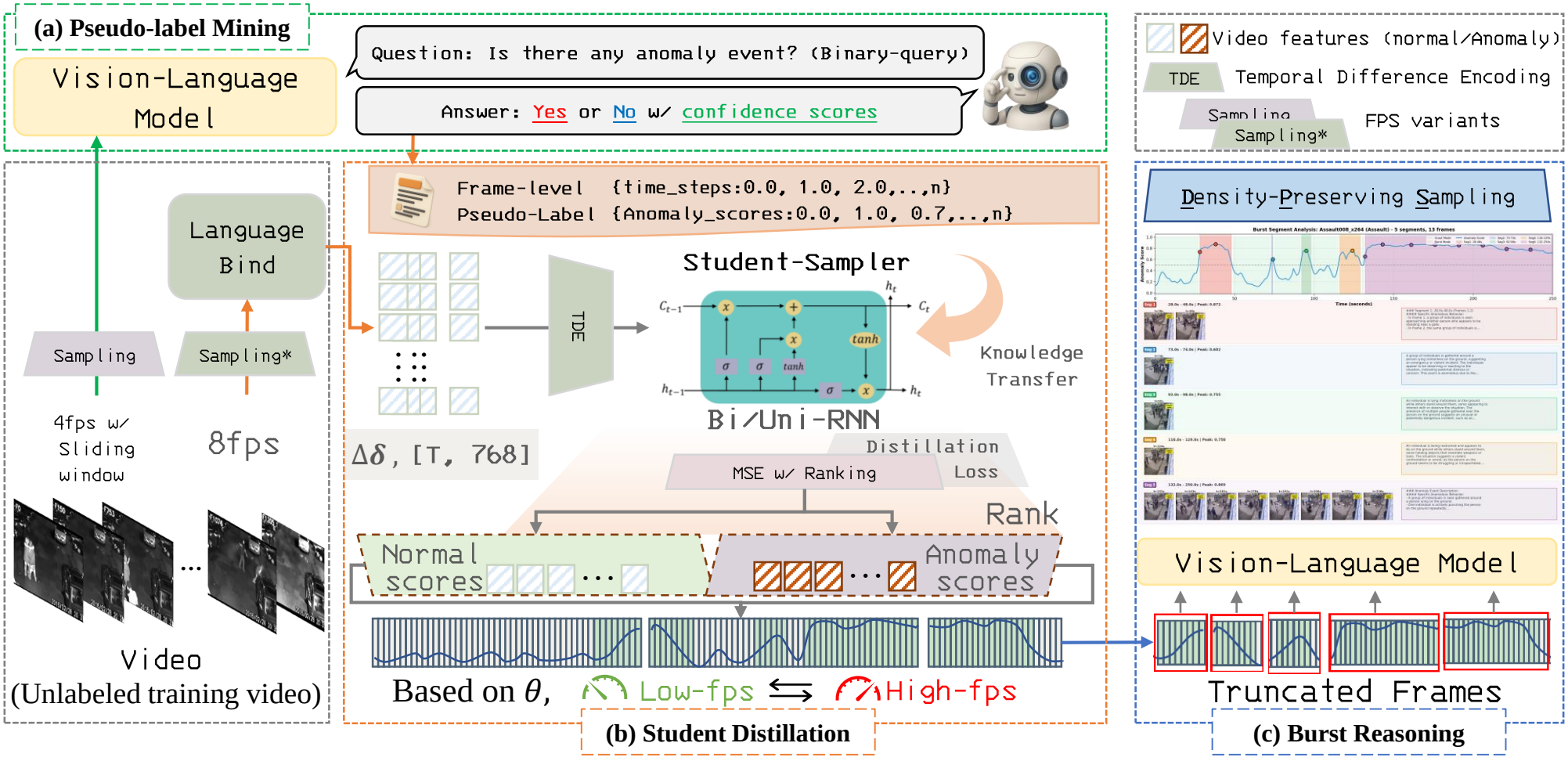

(a) Pseudo-label Mining
Vision-Language Model
Question: Is there any anomaly event? (Binary-query)
Answer: Yes or No w/ confidence scores
Video features (normal/Anomaly)
TDE
Temporal Difference Encoding
Sampling
Sampling*
FPS variants
Language Bind
Sampling
Sampling*
4fps w/ Sliding window
8fps
Video
(Unlabeled training video)
Frame-level {time_steps:0.0, 1.0, 2.0,..,n}
Pseudo-Label {Anomaly_scores:0.0, 1.0, 0.7,..,n}
TDE
Student-Sampler
Knowledge Transfer
Bi/Uni-RNN
ΔΔ, [T, 768]
Distillation Loss
MSE w/ Ranking
Rank
Normal scores
Anomaly scores
Based on θ,
Low-fps
High-fps
(b) Student Distillation
Density-Preserving Sampling
Vision-Language Model
Truncated Frames
(c) Burst Reasoning